

Transformations of Functions

- 1) In the multiple-choice questions 1 through 4, the solid graph shows $f(x)$. Identify which function is represented by the dashed graph.
- 2) Identify which function is represented by the dashed graph.
- 3) Identify which function is represented by the dashed graph.
- 4) Identify which function is represented by the dashed graph.
- 5) Describe the transformation of $g(x) = (x + 4)^2$ from its parent graph.
- 6) Describe the transformation if $f(x) = x^2$ is replaced with $2f(x)$.
- 7) Given $g(x) = f(x - 1) - 5$, describe the transformation to $f(x)$ that results in $g(x)$.
- 8) Describe the transformation of $f(x) = \frac{1}{2}(x - 3)^2$ from its parent graph.
- 9) Given $g(x) = f(x - 1) - 5$, describe the transformation to $f(x)$ that results in $g(x)$.
- 10) Describe the transformation of $f(x) = \frac{1}{2}(x - 3)^2$ from its parent graph.
- 11) Given the graph of f shown, sketch $y = -f(x + 3) - 2$ and $y = 3f(x - 1)$.
- 12) Given the graph of g shown, sketch $y = -g(x - 2)^2$ and $y = 2g(x) + 3$.
- 13) Given the graph of $f(x)$ shown, sketch $y = f(2x)$ and $y = f(-x) + 2$.
- 14) Given that the point $(1, 2)$ lies on $y = f(x)$, find the corresponding point on $y = 3f(2x)$.
- 15) Identify the coordinates of the point $(3, -2)$ translated 5 units left and 6 units up.
- 16) Given parent function $f(x) = x^3$, write the transformed equation after shifting left 4, shifting up 6, reflecting over the x -axis, and vertically shrinking by $\frac{1}{3}$.
- 17) Given parent function $f(x) = \sqrt{x}$, write the transformed equation after shifting left 5, shifting up 7, reflecting over the x -axis and y -axis, and vertically stretching by 2.
- 18) Given parent function $f(x) = \sqrt{x}$, write the transformed equation after shifting right 12, shifting down 5, reflecting over the y -axis, and horizontally stretching by 2.
- 19) The graph of $y = x^2$ may be transformed into $y = 5 - 3(x - 4)^2$ by a reflection in the line $y = 0$, a vertical stretch with scale factor k , a horizontal translation of p units, and a vertical translation of q units. Determine k , p , and q .